Vinay Kumar Mandadi

AI and ML Engineer

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SUMMARY

Computer Science graduate (LPU, 2026) focused on reliable ML: sole-author research on model failure under dataset shift and calibration, multi-agent LLM validation tools, and hands-on AI evaluation work.

SKILLS

ML: Python, scikit-learn, pandas, NumPy, PyTorch, TensorFlow, Random Forest, SMOTE, SHAP, Platt scaling and calibration

LLM systems: Multi-agent pipelines, Gemini and OpenAI APIs, Tavily, MCP, prompt-injection defences, evaluation

Deep learning: EfficientNetV2, BiLSTM, attention, Grad-CAM, ONNX

Serving and infra: FastAPI, Flask, Docker, Kubernetes, Streamlit, Oracle Cloud, Google Cloud

RESEARCH

Explainable and Trustworthy Network Intrusion Detection: From Multi-Class SHAP Analysis to Confidence-Aware Deployment

Sole author. Accepted at IEEE IS'26, 13th International Conference on Intelligent Systems; not presented. Preprint: doi.org/10.5281/zenodo.21260892.

99.02% accuracy (macro F1 0.906) on CIC-IDS2017, collapsing to 31.94% with 0% attack recall on UNSW-NB15 without retraining. Confidence filtering fails under this shift; Platt scaling restores 68.0% out-of-domain accuracy (ECE 0.622 to 0.097).

AIRIMF: An AI-Driven Risk Identification and Mitigation Framework for Software Project Management

Sole author. Preprint: doi.org/10.5281/zenodo.20691021.

Predicts high-risk issues the moment a ticket is filed: 76.21% cross-validated accuracy and 82% high-risk recall on VS Code (2,593 bug reports), 83.37% and 89% on React (744); significantly better than SVM (McNemar, $p < 0.0001$).

Combating Synthetic Media Manipulation Using Deep Learning-Based Deepfake Detection Systems

Co-author (five LPU students with supervisor Harjit Kaur). Accepted at 8BIC 2026.

Proposes AFDDA, which combines spatial CNN features with temporal frame-difference features into a weighted deepfake artifact score.

PROJECTS

Veridian and AMAV — Multi-agent LLM validation

Python · FastAPI · Gemini · GPT-4o-mini · Tavily · MCP · Kubernetes · github.com/vinaykumarchowdary18/veridian

- Veridian (Kaggle AI Agents Intensive Capstone 2026, Agents for Business): an analyst model drafts an answer from live retrieved evidence, two auditors from different model families critique it, and an arbiter outputs a consensus score and a return, revise or escalate decision.
- Treats retrieved web content as untrusted: prompt-injection screening, input bounds, rate limiting, audit logging. Predecessor AMAV runs containerised on Kubernetes.

EXPERIENCE

AI Evaluation Contributor (freelance), Handshake AI — Remote · Jul 2026 – Present

- Write hard, multi-step research questions that current AI models get wrong, with answer trails verified against independent primary sources, used to evaluate and train AI systems. All submitted tasks approved on human review.

EDUCATION

Lovely Professional University (LPU), Punjab, India — Aug 2022 – Jul 2026

B.Tech, Computer Science and Engineering (specialisation: Google Cloud Computing)

Capstone: Truth in Pixels: Advancing Deep Fake Models for Reliable Deepfake Detection — Jan – May 2026

Team of five, supervised by Harjit Kaur. EfficientNetV2-M, BiLSTM and attention fused with a DCT frequency branch; AUC-ROC 0.968 to 0.992 on FaceForensics++ (C23), 91.4% to 93.1% accuracy on Celeb-DF v2 (cross-dataset); Grad-CAM, ONNX export, Flask app.

ADDITIONAL

Certifications: Oracle Cloud Infrastructure 2025 Multicloud Architect Professional, DevOps Professional and AI Foundations Associate (Oct 2025); Oracle Data Platform 2025 Foundations Associate (May 2026)

Languages: English (professional), Telugu (native), Hindi, Japanese (elementary, studying toward JLPT N2)